Name:I.Harshika
Roll No:2520030607

Skill-2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

Step:1

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo apt update
[sudo] password for harshika:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1193 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1449 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [238 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [315 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1711 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1228 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [344 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1883 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [248 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1788 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [430 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [411 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [45.4 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [12.4 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [50.7 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [14.3 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5744 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Fetched 12.5 MB in 7s (1837 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
174 packages can be upgraded. Run 'apt list --upgradable' to see them.
harshika@Harshika:~/CNNP-Skill2$ sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
iproute2 is already the newest version (6.1.0-1ubuntu6.4).
iproute2 set to manually installed.
```

Step:2

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns add host1
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns add router
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns add host2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns list
host2
router
host1
harshika@Harshika:~/CNNP-Skill2$
```

Step:3

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip link add veth-h1 type veth peer name veth-r1
harshika@Harshika:~/CNNP-Skill2$ sudo ip link add veth-r2 type veth peer name veth-h2
```

Step:4

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip link set veth-h1 netns host1
harshika@Harshika:~/CNNP-Skill2$ sudo ip link set veth-r1 netns router
harshika@Harshika:~/CNNP-Skill2$ sudo ip link set veth-r2 netns router
harshika@Harshika:~/CNNP-Skill2$ sudo ip link set veth-h2 netns host2
```

Step:5.1

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 ip link set veth-h1 up
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 ip link set lo up
```

Step:5.2

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router ip link set veth-r1 up
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router ip link set veth-r2 up
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router ip link set lo up
```

Step:5.3

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 ip link set veth-h2 up
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 ip link set lo up
```

Step:6

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
harshika@Harshika:~/CNNP-Skill2$
```

Step:7

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.236 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.057 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.070 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2026ms
rtt min/avg/max/mdev = 0.057/0.121/0.236/0.081 ms
```

Step:8

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1%
```

Step:10.1

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
1
```

Step:10.2

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
2
```

Step:10.3

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
1
```

Step:11

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 nc -l 1234
[sudo] password for harshika:
```

Step:12.1

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo] password for harshika:
Hello from host1!
This is my CNNP Skill Lab
Three Point Capture Lab
Message going through router
```

Step:12.2

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 nc -l 1234
[sudo] password for harshika:
Hello from host1!
This is my CNNP Skill Lab
Three Point Capture Lab
Message going through router
```

Step:12.3

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
22
```

Step:12.4

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
23
```

Step:12.5

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
[sudo] password for harshika:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
23
```

Step:13

```
harshika@Harshika:~$ cd CNNP-Skill2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host2 nc -l 1234
[sudo] password for harshika:
Hello from host1!
This is my CNNP Skill Lab
Three Point Capture Lab
Message going through router
^C
harshika@Harshika:~/CNNP-Skill2$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
harshika@Harshika:~/CNNP-Skill2$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.6K Nov 12 08:16 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.6K Nov 12 08:16 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.5K Nov 12 08:16 /tmp/cap3.pcap
harshika@Harshika:~/CNNP-Skill2$
```

Step:14.1

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
```

Step:14.2

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
harshika@Harshika:~/CNNP-Skill2$
```

Step:14.3

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      5 64
```

Step:15

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a35 3436 3836 3937  431210a..5468697
00000030: 3332 3036 3937 3332 3036 6437 3932 3034  3206973206d79204
00000040: 3334 6534 6535 3032 3035 3336 6236 3936  34e4e5020536b696
00000050: 6336 6332 3034 6336 3136 3230 610a 0a35  c6c204c61620a..5
00000060: 3436 3837 3236 3536 3532 3035 3036 6636  46872656520506f6
00000070: 3936 6537 3432 3034 3336 3137 3037 3437  96e7420436170747
00000080: 3537 3236 3532 3034 6336 3136 3230 610a  57265204c61620a.
00000090: 0a34 6436 3537 3337 3336 3136 3736 3532  .4d6573736167652
000000a0: 3036 3736 6636 3936 6536 3732 3037 3436  0676f696e6720746
000000b0: 3837 3236 6637 3536 3736 3832 3037 3236  8726f75676820726
000000c0: 6637 3537 3436 3537 3230 610a 0a         f757465720a..
```

Step:16.1

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
468.040369787
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
468.040377131
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
402.524501152
```

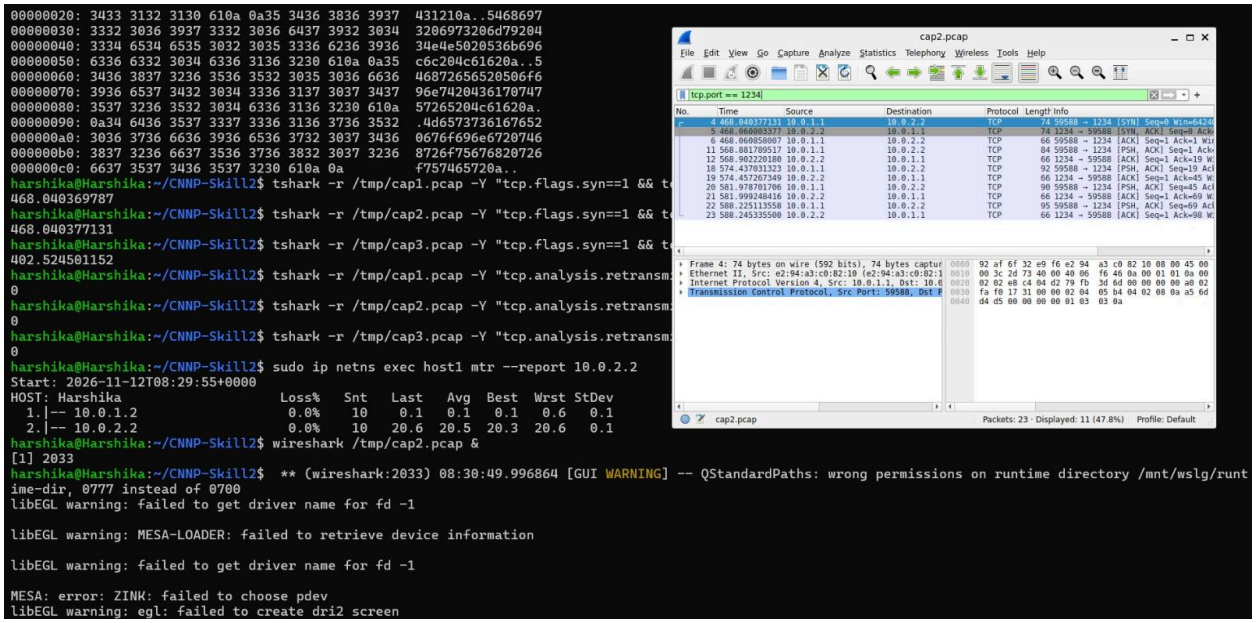
Step:16.2

```
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
harshika@Harshika:~/CNNP-Skill2$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
```

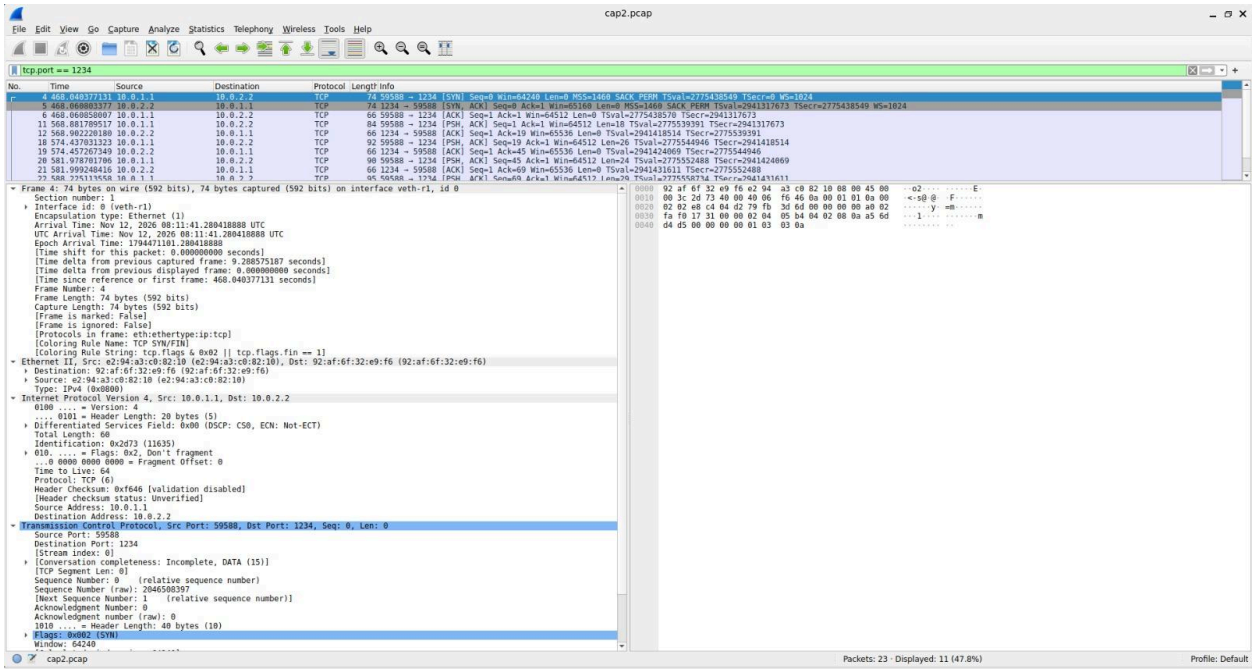
Step:17

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T08:29:55+0000
HOST: Harshika                    Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.1   0.1   0.6   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.6  20.5  20.3  20.6   0.1
```

Step:18.1



Step:18.2



Step:19

```
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns del host1
Cannot remove namespace file "/run/netns/host1": No such file or directory
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns del router
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns del host2
harshika@Harshika:~/CNNP-Skill2$ sudo ip netns list
```